

=====ОБЩЕЕ=====

ИМЯ – ТАБЛИЦА!

```
hostnamectl set-hostname isp.sirius-exam.org; exec bash
(# isp/hq-srv/hq-cli/br-srv .sirius-exam.org)
```

ИНТЕРНЕТ

```
nmtui
ip firewall disable
sysctl net.ipv4.ip_forward=1 >> /etc/sysctl.conf
sysctl -p
```

ПРОВЕРКА

```
ip -c -br a
```

```
sudo nano /etc/selinux/config
SELINUX=disabled
setenforce 0
getenforce – проверка что отключен
```

ВРЕМЯ

```
timedatectl set-timezone Europe/Moscow
```

DNS (потом исправить)

```
sudo nano /etc/systemd/resolved.conf
[Resolve]
DNS=192.168.100.2 77.88.8.7
FallbackDNS=77.88.8.3
Domains=sirius-exam.org
```

```
systemctl restart systemd-resolved
```

=====ISP=====

УСТАНОВКА

```
dnf install -y nftables nginx httpd-tools chrony
```

-----ПРОБРОС ПОРТОВ/ИНТЕРНЕТ/DNS

```
nano /etc/rc.d/rc.local
ip route add 192.168.100.0/27 via 172.16.1.2 (BR)
ip route add 192.168.200.0/28 via 172.16.1.2 (BR)
ip route add 192.168.30.0/28 via 172.16.2.2 (HQ)
```

```
chmod +x /etc/rc.d/rc.local
nmcli connection up ens19 && nmcli connection up ens20 (на всякий)
```

NFTABLES

nano /etc/nftables/isp.nft

```
table inet nat {
    chain POSTROUTING {
        type nat hook postrouting priority srcnat;
        oifname "ens18_П0PT" masquerade
    }
}
```

nano /etc/sysconfig/nftables.conf

```
include "/etc/nftables/isp.nft"
```

systemctl enable --now nftables && systemctl restart nftables

ПР0БЕРКА

sudo nft list ruleset

ping ya.ru

-----УЧЕТКА

useradd net_admin

echo 'net_admin:P@ssw0rd' | chpasswd

usermod -aG wheel net_admin

nano /etc/sudoers.d/net_admin

```
net_admin ALL=(ALL) NOPASSWD: ALL
```

chmod 440 /etc/sudoers.d/net_admin

-----CHRONY

nano /etc/chrony.conf

```
server ntp1.vniiftri.ru iburst prefer
local stratum 5
allow 0.0.0.0/0
driftfile /var/lib/chrony/drift
makestep 1.0 3
rtcsync
logdir /var/log/chrony
```

systemctl enable --now chronyd && systemctl restart chronyd

-----Nginx reverse proxy

```

nano /etc/nginx/conf.d/web.conf
server {
    listen 80;
    server_name web.sirius-exam.org;
    location / {
        proxy_pass http://172.16.1.2:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Real-IP $remote_addr;
        auth_basic "Restricted area";
        auth_basic_user_file /etc/nginx/.htpasswd;
    }
}

```

```

nano /etc/nginx/conf.d/docker.conf
server {
    listen 80;
    server_name docker.sirius-exam.org;
    location / {
        proxy_pass http://172.16.2.2:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Real-IP $remote_addr;
    }
}

```

SSL для Nextcloud:

```

mkdir -p /etc/nginx/ssl
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout
/etc/nginx/ssl/nextcloud.key -out /etc/nginx/ssl/nextcloud.crt -subj
"/C=RU/ST=Moscow/L=Moscow/O=Sirius/OU=IT/CN=nextcloud.sirius-exam.org"

```

```

nano /etc/nginx/conf.d/nextcloud.conf
server {
    listen 443 ssl http2;
    server_name nextcloud.sirius-exam.org;
    ssl_certificate /etc/nginx/ssl/nextcloud.crt;
    ssl_certificate_key /etc/nginx/ssl/nextcloud.key;
    client_max_body_size 512M;
    add_header Strict-Transport-Security "max-age=31536000;
includeSubDomains; preload" always;
    location / {
        proxy_pass http://172.16.2.2:8081;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}

```

```
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_buffering off;
        proxy_request_buffering off;
    }
}
```

```
-----htpasswd + zanyck
echo 'P@ssw0rd' | htpasswd -i -c /etc/nginx/.htpasswd WEB
nginx -t
systemctl enable --now nginx
systemctl restart nginx
```

```
=====HQ-SRV=====
# УСТАHOВКА
dnf install -y bind bind-utils chrony mdadm nfs-utils nfs4-acl-tools httpd
php php-mysqld mariadb-server mariadb
```

```
-----УЧЕТКИ
useradd -u 2026 sshuser
usermod -aG wheel sshuser
echo 'sshuser:P@ssw0rd' | chpasswd
echo 'sshuser ALL=(ALL) NOPASSWD: ALL' > /etc/sudoers.d/sshuser
chmod 440 /etc/sudoers.d/sshuser
```

```
# SSH
nano /etc/ssh/sshd_config
    Port 2026
    MaxAuthTries 2
    AllowUsers sshuser
    PasswordAuthentication yes
    Banner /etc/ssh/banner

echo 'Authorized access only' > /etc/ssh/banner
systemctl restart sshd
semanage port -a -t ssh_port_t -p tcp 2026
systemctl restart sshd
```

```
-----Chrony (клиент)
nano /etc/chrony.conf
    server 172.16.1.1
    iburst driftfile /var/lib/chrony/drift
    makestep 1.0 3
    rtcsync
    logdir /var/log/chrony

systemctl enable --now chronyd `systemctl restart chronyd`
```

-----DNS

nano /etc/named.conf

```
options {
    listen-on port 53 { any; };
    listen-on-v6 port 53 { none; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { any; };
    recursion yes;
    dnssec-validation no;
    forward first;
    forwarders { 77.88.8.7; };
    managed-keys-directory "/var/named/dynamic";
    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";
};
logging { channel default_debug { file "data/named.run"; severity dynamic;
}; };
zone "." IN { type hint; file "named.ca"; };
zone "sirius-exam.org" { type master; file "master/sirius-exam.org"; };
zone "100.168.192.in-addr.arpa" { type master; file "master/100.168.192.in-
addr.arpa"; };
zone "200.168.192.in-addr.arpa" { type master; file "master/200.168.192.in-
addr.arpa"; };
zone "30.168.192.in-addr.arpa" { type master; file "master/30.168.192.in-
addr.arpa"; };
include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
```

Проверка

named-checkconf

ДИРЕКТОРИЯ ДЛЯ ЗОН

mkdir -p /var/named/master

chown root:named /var/named/master

chmod 750 /var/named/master

```

# Создание зон
nano /var/named/master/sirius-exam.org
$TTL 604800
sirius-exam.org. IN SOA ns01.sirius-exam.org. root.sirius-exam.org.
(2026052801 600 3600 1w 360)
    IN NS ns01.sirius-exam.org.
ns01      IN A   192.168.100.2
hq-srv    IN A   192.168.100.2
hq-rtr    IN A   192.168.100.1
hq-cli    IN A   192.168.200.2
br-srv    IN A   192.168.30.2
br-rtr    IN A   192.168.30.1
isp       IN A   172.16.1.1
web       IN CNAME isp.sirius-exam.org.
docker    IN CNAME isp.sirius-exam.org.

-----
nano /var/named/master/100.168.192.in-addr.arpa
$TTL 604800
100.168.192.in-addr.arpa. IN SOA ns01.sirius-exam.org. root.sirius-exam.org.
(2026052801 600 3600 1w 360)
    IN NS ns01.sirius-exam.org.
2      IN PTR hq-srv.sirius-exam.org.
1      IN PTR hq-rtr.sirius-exam.org.

-----
nano /var/named/master/200.168.192.in-addr.arpa
$TTL 604800
200.168.192.in-addr.arpa. IN SOA ns01.sirius-exam.org. root.sirius-exam.org.
(2026052801 600 3600 1w 360)
    IN NS ns01.sirius-exam.org.
2      IN PTR hq-cli.sirius-exam.org.
1      IN PTR hq-rtr.sirius-exam.org.

-----
nano /var/named/master/30.168.192.in-addr.arpa
$TTL 604800
30.168.192.in-addr.arpa. IN SOA ns01.sirius-exam.org. root.sirius-exam.org.
(2026052801 600 3600 1w 360)
    IN NS ns01.sirius-exam.org.
2      IN PTR br-srv.sirius-exam.org.
1      IN PTR br-rtr.sirius-exam.org.

-----
chown root:named /var/named/master/* && chmod 640 /var/named/master/*
systemctl enable --now named && systemctl restart named

```

После запуска BIND – перенастроить resolved на HQ-SRV

```
nano /etc/systemd/resolved.conf.d/dns.conf
```

```
[Resolve]
```

```
DNS=127.0.0.1
```

```
FallbackDNS=77.88.8.7
```

```
Domains=sirius-exam.org
```

```
systemctl restart systemd-resolved
```

Прописать в DNS свой же адрес

BR-SRV прописать в DNS HQ-SRV

Проверка

```
named-checkconf -z
```

```
ping au-team.irpo
```

```
host hq-rtr.au-team.irpo
```

```
host 192.168.100.1
```

-----RAID 0

Определить диски: lsblk (sdb+sdc или vdb+vdc)

```
mdadm --create /dev/md0 --level=0 --raid-devices=2 /dev/sdb /dev/sdc --run
```

```
mdadm --detail --scan >> /etc/mdadm.conf
```

```
mkfs.ext4 /dev/md0
```

```
mkdir -p /raid
```

```
mount /dev/md0 /raid
```

```
echo '/dev/md0 /raid ext4 defaults 0 2' >> /etc/fstab
```

-----NFS-сервер

```
mkdir -p /raid/nfs
```

```
chmod 777 /raid/nfs
```

```
nano /etc/exports
```

```
/raid/nfs 192.168.200.0/28(rw,no_root_squash)
```

```
systemctl enable --now nfs-server && systemctl restart nfs-server &&
```

```
exportfs -arv
```

-----Apache + MariaDB (веб-приложение)

```
mkdir -p /mnt/cdrom
```

```
mount -o ro /dev/sr0 /mnt/cdrom
```

```
cp -r /mnt/cdrom/web/ /home/user/web/
```

```
systemctl enable --now mariadb
```

```
mysql -u root -e "ALTER USER 'root'@'localhost' IDENTIFIED BY 'P@ssw0rd';"
```

```
mysql -u root -pP@ssw0rd -e "DELETE FROM mysql.user WHERE User=''; DELETE
FROM mysql.user WHERE User='root' AND Host NOT IN
('localhost','127.0.0.1','::1'); DROP DATABASE IF EXISTS test; FLUSH
PRIVILEGES;"
mysql -u root -pP@ssw0rd -e "CREATE DATABASE IF NOT EXISTS webdb; CREATE
USER IF NOT EXISTS 'web'@'%' IDENTIFIED BY 'P@ssw0rd'; GRANT ALL PRIVILEGES
ON webdb.* TO 'web'@'%'; FLUSH PRIVILEGES;"
mysql -u root -pP@ssw0rd webdb < /home/user/web/dump.sql

cp /home/user/web/index.php /home/user/web/logo.png /var/www/html/
chown -R apache:apache /var/www/html/
```

Отредактировать `nano /var/www/html/index.php` – найти и заменить параметры подключения к БД:

```
$servername = "localhost";
$username = "web";
$password = "P@ssw0rd";
$dbname = "webdb";
```

```
systemctl enable --now httpd && systemctl restart httpd
```

=====BR-SRV=====

УСТАНОВКА

```
dnf install -y chrony docker-ce docker-ce-cli docker-compose-plugin docker-
compose samba nginx openssl
```

-----УЧЕТКИ

```
useradd -u 2026 sshuser
usermod -aG wheel sshuser
echo 'sshuser:P@ssw0rd' | chpasswd
echo 'sshuser ALL=(ALL) NOPASSWD: ALL' > /etc/sudoers.d/sshuser
chmod 440 /etc/sudoers.d/sshuser
```

SSH

```
nano /etc/ssh/sshd_config
    Port 2026
    MaxAuthTries 2
    AllowUsers sshuser
    PasswordAuthentication yes
    Banner /etc/ssh/banner
```

```
echo 'Authorized access only' > /etc/ssh/banner
systemctl restart sshd
```



```
semanage port -a -t ssh_port_t -p tcp 2026
systemctl restart sshd
```

-----Chrony (клиент)

```
nano /etc/chrony.conf
server 172.16.1.1
iburst driftfile /var/lib/chrony/drift
makestep 1.0 3
rtcsync
logdir /var/log/chrony
```

```
systemctl enable --now chronyd `systemctl restart chronyd
```

-----Монтирование CD

```
mkdir -p /mnt/cdrom
mount -o ro /dev/sr0 /mnt/cdrom
cp -r /mnt/cdrom/docker/ /home/user/docker/
```

-----SAMBA

```
groupadd hq
useradd -G hq hquser1
echo -e "P@ssw0rd\nP@ssw0rd" | smbpasswd -a -s hquser1
useradd -G hq hquser2
echo -e "P@ssw0rd\nP@ssw0rd" | smbpasswd -a -s hquser2
useradd -G hq hquser3
echo -e "P@ssw0rd\nP@ssw0rd" | smbpasswd -a -s hquser3
useradd -G hq hquser4
echo -e "P@ssw0rd\nP@ssw0rd" | smbpasswd -a -s hquser4
useradd -G hq hquser5
echo -e "P@ssw0rd\nP@ssw0rd" | smbpasswd -a -s hquser5
```

```
nano /etc/samba/smb.conf
[global]
workgroup = SIRIUSEXAM
server string = BR-SRV Samba Server
netbios name = BR-SRV
security = user
map to guest = Bad User
dns proxy = no
realm = SIRIUS-EXAM.ORG
```

```
systemctl enable --now smb nmb && systemctl restart smb nmb
```

```
-----Docker – загрузка образов
systemctl enable --now docker
docker load < /home/user/docker/site_latest.tar
docker load < /home/user/docker/mariadb_latest.tar
```

Docker – Testapp

nano /root/web.yaml – (отдельный файл)

cd /root && docker-compose -f web.yaml up -d

Подождать ~30 сек, затем:

```
docker exec db mysql -u root -pP@ssw0rd -e "CREATE DATABASE IF NOT EXISTS
testdb; CREATE USER IF NOT EXISTS 'test'@'%' IDENTIFIED BY 'P@ssw0rd'; GRANT
ALL PRIVILEGES ON testdb.* TO 'test'@'%; FLUSH PRIVILEGES;"
docker-compose -f web.yaml restart
```

Docker – Nextcloud

mkdir -p /opt/nextcloud/db_data /opt/nextcloud/nextcloud_data

nano /opt/nextcloud/docker-compose.yml (Отдельный файл)

cd /opt/nextcloud

docker-compose up -d

-----Nginx + SSL на BR-SRV (для Nextcloud)

```
mkdir -p /etc/nginx/ssl /etc/nginx/sites-available /etc/nginx/sites-enabled
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout
/etc/nginx/ssl/nextcloud.key -out /etc/nginx/ssl/nextcloud.crt -subj
"/C=RU/ST=Moscow/L=Moscow/O=Sirius/OU=IT/CN=127.0.0.1"
```

nano /etc/nginx/sites-available/nextcloud.conf

```
server {
    listen 443 ssl http2;
    server_name 127.0.0.1;
    ssl_certificate /etc/nginx/ssl/nextcloud.crt;
    ssl_certificate_key /etc/nginx/ssl/nextcloud.key;
    client_max_body_size 512M;
    add_header Strict-Transport-Security "max-age=31536000;
includeSubDomains; preload" always;
    location / {
        proxy_pass http://127.0.0.1:8081;
        proxy_set_header Host $host;
```

```
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_buffering off;
    proxy_request_buffering off;
}
}
```

```
ln -sf /etc/nginx/sites-available/nextcloud.conf /etc/nginx/sites-enabled/nextcloud.conf
```

В /etc/nginx/nginx.conf добавить в блок http: include /etc/nginx/sites-enabled/*.conf;

```
rm -f /etc/nginx/conf.d/default.conf
```

```
nginx -t && systemctl enable --now nginx && systemctl restart nginx
```

=====HQ-CLI=====

CLI – не трогай у него DHCP

УСТАНОВКА

```
dnf install -y chrony nfs-utils samba-client
```

-----УЧЕТКИ

```
useradd -u 2026 sshuser
```

```
usermod -aG wheel sshuser
```

```
echo 'sshuser:P@ssw0rd' | chpasswd
```

```
echo 'sshuser ALL=(ALL) NOPASSWD: ALL' > /etc/sudoers.d/sshuser
```

```
chmod 440 /etc/sudoers.d/sshuser
```

SSH

```
nano /etc/ssh/sshd_config
```

```
Port 2026
```

```
MaxAuthTries 2
```

```
AllowUsers sshuser
```

```
PasswordAuthentication yes
```

```
Banner /etc/ssh/banner
```

```
echo 'Authorized access only' > /etc/ssh/banner
```

```
systemctl restart sshd
```

```
semanage port -a -t ssh_port_t -p tcp 2026
```

```
systemctl restart sshd
```

-----Chrony (клиент)

```
nano /etc/chrony.conf
```

```
server 172.16.1.1
iburst driftfile /var/lib/chrony/drift
makestep 1.0 3
rtcsync
logdir /var/log/chrony
```

```
systemctl enable --now chronyd` `systemctl restart chronyd
```

-----NFS-КЛИЕНТ

```
mkdir -p /mnt/nfs && chmod 777 /mnt/nfs
echo '192.168.100.2:/raid/nfs /mnt/nfs nfs auto 0 0' >> /etc/fstab
mount -a
```

-----Пользователи и группы (Мод2)

```
groupadd Work
groupadd Job
groupadd labor
useradd -G Work User1
useradd -G Work User2
useradd -G Job User3
useradd -G Job User4
useradd -G labor User5
mkdir -p /home/Folder
mkdir /home/Folder/work_shared
chown root:Work /home/Folder/work_shared
chmod 770 /home/Folder/work_shared
mkdir /home/Folder/job_readonly
chown root:Job /home/Folder/job_readonly
chmod 750 /home/Folder/job_readonly
```

Ошибки

1. semanage: команда не найдена
dnf install policycoreutils-python -y